

```

1
*****
*****
1 ! *****
2
3
*****
*****
3 ! *****
4
5
*****
*****
5 ! *****
6 20May09 When subdividing activity codes into pseudo codes to the
CARMM program, these are the
6 ! places where they also
7 will need to be added:
8 Below where %addActvCodeToMxMail is being
called in (appends the
8 ! psuedo code to the mix mail map)
9 Below where Proc format is formatting all
the activity codes
10 In the excel file (ActivityCodesMaster.xls)
11 In the sas code of "mapclassRecast.sas"
NOTE not needed for delivery
11 ! cost model, as not
12 using DMM field.
13 In the mixed mail/direct actv code match file
"MxMailCode.csv" - add new activity
13 ! codes for Std Mail
14 Be sure to use the correct field in ActivityCodeMaster -
NewCRAProd
15 select KL.*, ac.NewCRAProd as class2
16
*****
*****
16 ! *****
17 REMOVE COMMENT ON SELECT CASING DEFINITION SECTION TO USE FOR CASING
ONLY!!!
18
*****
*****
18 ! *****;
19 resetline;
1 options MPRINT SYMBOLGEN;
2
3
4 libname IOCS 'D:\ExternalSSD\Folder 19\FY25\SAS\SASLib';
NOTE: Libref IOCS was successfully assigned as follows:
Engine: V9
Physical Name: D:\ExternalSSD\Folder 19\FY25\SAS\SASLib
5 %let pathProg = D:\ExternalSSD\Folder 19\FY25\SAS\SASInputFiles;
6 %let pathOut = D:\ExternalSSD\Folder 19\FY25\SAS\KLTables;

```

```

7      %let FY = 25;
8      %let FYData = 25;
9
10     %macro assignRouteGroup();
11         *** The following statement assigns route code groups
12         *** so that reports can be created separately for
13         *** these groups.;
14
15         *set rgroup for special runs, for delivery cost model (3
values);
16         *Put Letter Route types in RGroup 1, SPR in RGroup 2, 99 in
RGroup 3;
17
18         if '71' <= route <= '83' then RGroup=1;
19         else if '84' <= route <= '98' then RGroup=2;
20         else RGroup=3;
21
22     %mend;
23
24
25     data b;
26     set IOCS.PRCPubCL25;
27     title1 'FY 20&FY. Qtr A CARMM System';
28     /***** The following variables are updated using Cluster
dataset
28 ! *****/
29     /***** Updated by Dan Zhang on Nov 16, 2020
29 ! *****/
30     dollar = READCOST; /* Cost (in dollar) for each carrier
reading based on cluster
30 ! sampling method */
31     act1st = ' ';
32     act234 = ' ';
33     F9252 = CRAFTGRP; /* Craft Subscript: 5=Full-Time Carrier;
6=Part-Time Carrier */
34     f264 = WEIGHTCAG; /* CAG Group used in the Control Total
Dollar: A-H */
35
/*****
*****
35 ! *****/
36
37     *** This section is used for Folder 19, Delivery Cost Model
37 ! ****;
38     *** Subdivide ECR HD/SAT into DAL and non-DAL for use in Folder
19 Delivery Cost Model**;
39     IF F262 in ('1317', '2317', '3317', '4317') then do;
40         act1st = substr(F262,1,1);
41         act234 = substr(F262,2,3);
42         if Q23E10 = 'N' then act234 = '319'; ** Parent pc w/ no
DAL;
43         else
44         if (Q23J02 = 'B' or

```

```

45          Q23H03      = 'H') then do; ** WSH or MMS=EH marking eff
FY25Q1;
46          /* (Q23J02      = 'A' or
47          Q23H03      = 'H') then do; ** WSH or MMS=EH marking; */
48
49          if (q23B01      = 'Y' and substr(F262,1,1) ne '1')
50              then act234 = '315'; ** DAL, for non-letters;
51              else act234 = '317';
52          end;
53          else
54          if (Q23J02      = 'A' or
55          Q23J02      = 'C' or
56          Q23H03      = 'I') then do; **WSS or MMS=ES marking eff
FY25Q1;
57          /* (Q23J02      = 'B' or
58          Q23H03      = 'I') then do; **WSS or MMS=ES marking; */
59          if (Q23B01      = 'Y' and substr(F262,1,1) ne '1')
60              then act234 = '316'; ** DAL, for non-letters;
61              else act234 = '318';
62          end;
63          F262 = act1st||act234;
64      END;
65
66          ** CARMM Edits;
67          if f264='K' then f9252='9';
68          if f261 ne '4';
69          if dollar=0 and q01=' ' then delete;
70          else do;
71              if f260 in ('84', '89') then f260='87';
72          end;
73          if f9252 in ('5','6'); **carrier craft;
74      run;

```

NOTE: Character values have been converted to numeric values at the places given by:

```

      (Line):(Column).
      67:32    73:12

```

NOTE: Variable q01 is uninitialized.

NOTE: There were 150303 observations read from the data set  
IOCS.PRCPUBCL25.

NOTE: The data set WORK.B has 102200 observations and 303 variables.

NOTE: DATA statement used (Total process time):

```

      real time          0.19 seconds
      cpu time           0.03 seconds

```

SYMBOLGEN: Macro variable PATHPROG resolves to D:\ExternalSSD\Folder  
19\FY25\SAS\SASInputFiles

```

75
76      *Read in file that maps activity codes into mixed mail codes,
77      and add all new activity codes added above;
78      %include "&pathProg\macMxMail_AMS.sas" / source2; /*GET MORE DETAILS
IN THE LOG FOR THIS
78 ! INCLUDE STATEMENT*/

```

```

NOTE: %INCLUDE (level 1) file D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\macMxMail_AMS.sas
      is file D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\macMxMail_AMS.sas.
79  +*Macros to read mixed mail activity code mapping, and insert new
activity codes;
80  +
81  +%macro readMxMail(CSVFile,MxMailMap);
82  +*Read file CSVFile (CSV format) with activity codes that belong to
mixed mail codes;
83  +* Prepare final mixed mail mapping dataset;
84  +Data mxmaill;
85  + infile &CSVfile DLM="," MISSOVER firstobs=2;
86  + format mixcode $4.;
87  + format dircode $4.;
88  + input mixcode $ dircode $;
89  +run;
90  +
91  +Data &MxMailMap; set MxMail1;
92  +run;
93  +%mend;
94  +
95  +%macro addActvCodeToMxMail(MxMailMap, actvMatch, actvNew);
96  +*insert new activity code actvNew with same mixed mail cosdes as
actvMatch;
97  +proc sql;
98  + insert into MxMailMap
99  + select mixcode, &actvNew
100 + from mxmaill
101 + where dircode = &actvMatch;
102 +quit;
103 +%mend;
104 +
105 +
106 +*Inserts activity code subsets that Nancy needs for processing,
these are not official
106!+activity codes;
107 +*AMS 17Nov17;
108 +%macro addActvCodeToMaster(ActivityCodesTable, ActivityCode,
NewCRAProd);
109 +*insert new activity code actvNew with same mixed mail cosdes as
actvMatch;
110 +proc sql;
111 + insert into ActivityCodesTable
112 + set ActivityCode = &ActivityCode.,
113 +       NewCRAProd = &NewCRAProd.;
114 +quit;
115 +%mend;
116 +
117 +
118 +/*
119 +%macro
addMixedMailShapeToMxMail(MxMailMailMap,mxMailNoShape,shapeCode);

```

```

120 +*Insert new mixed mail code with shape info, using first digit of
activity code
121 + as last digit of mixed mail code;
122 +proc sql;
123 +   insert into MxMailMap
124 +       select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
125 + from mxMail1
126 + where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) =
put(&shapeCode.,z1.);
127 +quit;
128 +%mend;*/
129 +%macro addMixedMailShapeToMxMail(MxMailMap,mxMailNoShape);
130 +*Insert new mixed mail code with shape info, using first digit of
activity code
131 + as last digit of mixed mail code;
132 +%let shapeCode = 1;
133 +proc sql;
134 +   insert into MxMailMap
135 +       select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
136 + from mxMail1
137 + where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) =
put(&shapeCode.,z1.);
138 +%let shapeCode = 2;
139 +proc sql;
140 +   insert into MxMailMap
141 +       select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
142 + from mxMail1
143 + where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) =
put(&shapeCode.,z1.);
144 +%let shapeCode = 3;
145 +proc sql;
146 +   insert into MxMailMap
147 +       select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
148 + from mxMail1
149 + where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) =
put(&shapeCode.,z1.);
150 +%let shapeCode = 4;
151 +proc sql;
152 +   insert into MxMailMap
153 +       select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
154 + from mxMail1
155 + where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) =
put(&shapeCode.,z1.);
156 +quit;
157 +%mend;
158 +
159 +
160 +/*
161 +*test macros;
162 +%let pathData = C:\AraSardar\IOCS\CARMM;
163 +%readMxMail("&pathData\MxMailCodeFY14.csv",MxMailMap)
164 +%addActvCodeToMxMail(MxMailMap,'1060','1061');
165 +%addActvCodeToMxMail(MxMailMap,'1060','1062');
166 +%addActvCodeToMxMail(MxMailMap,'1060','1063');

```

```

167 +%addActvCodeToMxMail (MxMailMap, '1310', '1311');
168 +%addActvCodeToMxMail (MxMailMap, '1310', '1312');
169 +%addActvCodeToMxMail (MxMailMap, '1310', '1313');
170 +%addActvCodeToMxMail (MxMailMap, '1310', '1314');
171 +%addActvCodeToMxMail (MxMailMap, '1310', '1315');
172 +%addActvCodeToMxMail (MxMailMap, '1310', '1316');
173 +%addActvCodeToMxMail (MxMailMap, '1310', '1317');
174 +%addActvCodeToMxMail (MxMailMap, '1310', '1318');
175 +%addActvCodeToMxMail (MxMailMap, '1310', '1319');
176 +%addActvCodeToMxMail (MxMailMap, '2310', '2311');
177 +%addActvCodeToMxMail (MxMailMap, '2310', '2312');
178 +%addActvCodeToMxMail (MxMailMap, '2310', '2313');
179 +%addActvCodeToMxMail (MxMailMap, '2310', '2314');
180 +%addActvCodeToMxMail (MxMailMap, '2310', '2315');
181 +%addActvCodeToMxMail (MxMailMap, '2310', '2316');
182 +%addActvCodeToMxMail (MxMailMap, '2310', '2317');
183 +%addActvCodeToMxMail (MxMailMap, '2310', '2318');
184 +%addActvCodeToMxMail (MxMailMap, '2310', '2319');
185 +%addActvCodeToMxMail (MxMailMap, '3310', '3311');
186 +%addActvCodeToMxMail (MxMailMap, '3310', '3312');
187 +%addActvCodeToMxMail (MxMailMap, '3310', '3313');
188 +%addActvCodeToMxMail (MxMailMap, '3310', '3314');
189 +%addActvCodeToMxMail (MxMailMap, '3310', '3315');
190 +%addActvCodeToMxMail (MxMailMap, '3310', '3316');
191 +%addActvCodeToMxMail (MxMailMap, '3310', '3317');
192 +%addActvCodeToMxMail (MxMailMap, '3310', '3318');
193 +%addActvCodeToMxMail (MxMailMap, '3310', '3319');
194 +%addActvCodeToMxMail (MxMailMap, '4310', '4311');
195 +%addActvCodeToMxMail (MxMailMap, '4310', '4312');
196 +%addActvCodeToMxMail (MxMailMap, '4310', '4313');
197 +%addActvCodeToMxMail (MxMailMap, '4310', '4314');
198 +%addActvCodeToMxMail (MxMailMap, '4310', '4315');
199 +%addActvCodeToMxMail (MxMailMap, '4310', '4316');
200 +%addActvCodeToMxMail (MxMailMap, '4310', '4317');
201 +%addActvCodeToMxMail (MxMailMap, '4310', '4318');
202 +%addActvCodeToMxMail (MxMailMap, '4310', '4319');
203 +proc sort data=MxMailMap nodup;
204 +by mixcode dircode;
205 +run;
206 +*/
207 +
208 +***** End macMxMail;
NOTE: %INCLUDE (level 1) ending.
SYMBOLGEN: Macro variable PATHPROG resolves to D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles
MPRINT(READMXMAIL): *Read file CSVFile (CSV format) with activity codes
that belong to mixed
mail codes;
MPRINT(READMXMAIL): * Prepare final mixed mail mapping dataset;
209 %readMxMail("&pathProg\MxMailCodeFY25.csv",MxMailMap); *for
domestic CRA;
MPRINT(READMXMAIL): Data mxmail1;
SYMBOLGEN: Macro variable CSVFILE resolves to "D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv"

```

```
MPRINT(READMXMAIL):   infile "D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv"
DLM="," MISOVER firstobs=2;
MPRINT(READMXMAIL):   format mixcode $4.;
MPRINT(READMXMAIL):   format dircode $4.;
MPRINT(READMXMAIL):   input mixcode $ dircode $;
MPRINT(READMXMAIL):   run;
```

```
NOTE: The infile "D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv" is:
      Filename=D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv,
      RECFM=V,LRECL=32767,File Size (bytes)=2624,
      Last Modified=13Nov2025:12:44:54,
      Create Time=17Nov2025:10:11:34
```

```
NOTE: 237 records were read from the infile "D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv".
      The minimum record length was 9.
      The maximum record length was 9.
NOTE: The data set WORK.MXMAIL1 has 237 observations and 2 variables.
NOTE: DATA statement used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds
```

```
SYMBOLGEN: Macro variable MXMAILMAP resolves to MxMailMap
MPRINT(READMXMAIL):   Data MxMailMap;
MPRINT(READMXMAIL):   set MxMail1;
MPRINT(READMXMAIL):   run;
```

```
NOTE: There were 237 observations read from the data set WORK.MXMAIL1.
NOTE: The data set WORK.MXMAILMAP has 237 observations and 2 variables.
NOTE: DATA statement used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds
```

```
210
MPRINT(ADDACTVCODETOMXMAIL):   *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
211 *add subdivided activity codes to mixed mail mappings (need to
update);
212
213 %addActvCodeToMxMail(MxMailMap,'1317','1315');
MPRINT(ADDACTVCODETOMXMAIL):   proc sql;
SYMBOLGEN: Macro variable ACTVNEW resolves to '1315'
SYMBOLGEN: Macro variable ACTVMATCH resolves to '1317'
MPRINT(ADDACTVCODETOMXMAIL):   insert into MxMailMap select mixcode,
'1315' from mxmail1 where
dircode = '1317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.03 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
214  %addActvCodeToMxMail(MxMailMap,'1317','1316');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1316'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'1316' from mxmaill where
dircode = '1317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.00 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
215  %addActvCodeToMxMail(MxMailMap,'1317','1318');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1318'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'1318' from mxmaill where
dircode = '1317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
216  %addActvCodeToMxMail(MxMailMap,'1317','1319');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1319'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'1319' from mxmaill where
dircode = '1317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.
```



```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
217  %addActvCodeToMxMail(MxMailMap,'2317','2315');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '2315'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '2317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'2315' from mxmaill where
dircode = '2317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.00 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
218  %addActvCodeToMxMail(MxMailMap,'2317','2316');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '2316'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '2317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'2316' from mxmaill where
dircode = '2317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.00 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
219  %addActvCodeToMxMail(MxMailMap,'2317','2318');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '2318'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '2317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'2318' from mxmaill where
dircode = '2317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.00 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
220 %addActvCodeToMxMail(MxMailMap,'2317','2319');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '2319'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '2317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'2319' from mxmaill where
dircode = '2317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
221 %addActvCodeToMxMail(MxMailMap,'3317','3315');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '3315'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '3317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'3315' from mxmaill where
dircode = '3317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
222 %addActvCodeToMxMail(MxMailMap,'3317','3316');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '3316'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '3317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'3316' from mxmaill where
dircode = '3317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
223 %addActvCodeToMxMail(MxMailMap,'3317','3318');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN: Macro variable ACTVNEW resolves to '3318'
SYMBOLGEN: Macro variable ACTVMATCH resolves to '3317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'3318' from mxmaill where
dircode = '3317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
224 %addActvCodeToMxMail(MxMailMap,'3317','3319');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN: Macro variable ACTVNEW resolves to '3319'
SYMBOLGEN: Macro variable ACTVMATCH resolves to '3317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'3319' from mxmaill where
dircode = '3317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
225 %addActvCodeToMxMail(MxMailMap,'4317','4315');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN: Macro variable ACTVNEW resolves to '4315'
SYMBOLGEN: Macro variable ACTVMATCH resolves to '4317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'4315' from mxmaill where
dircode = '4317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.00 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
226  %addActvCodeToMxMail(MxMailMap,'4317','4316');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '4316'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '4317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'4316' from mxmaill where
dircode = '4317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.00 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
227  %addActvCodeToMxMail(MxMailMap,'4317','4318');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '4318'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '4317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'4318' from mxmaill where
dircode = '4317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
228  %addActvCodeToMxMail(MxMailMap,'4317','4319');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '4319'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '4317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'4319' from mxmaill where
dircode = '4317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.00 seconds
      cpu time           0.00 seconds
```

```
229
230
231 proc sort data=MxMailMap nodup;
232         by dircode mixcode;
233 run;
```

```
NOTE: There were 293 observations read from the data set WORK.MXMAILMAP.
NOTE: 0 duplicate observations were deleted.
NOTE: The data set WORK.MXMAILMAP has 293 observations and 2 variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds
```

```
234
235 data carr; set b;
236     Craft=F9252;
237     Actv=F262;
238     *Actv=F9806; *mcz 10Aug08 use F9806 to distribute Int'l mixed
mail;
239     Route=F260;
240     BF=F261;
241     Fin=F263;
242     Cag=F264;
243     Cagfin=cag !! fin;
244
245     ****  SELECT CASING DEFINITION ***  ;
246     *To restrict to subset that is casing remove comment from next
line;
247     *if Q16F03A in ('A','B','C');  ** General casing (3 options) -- use
for Delivery Cost Model
247! inputs;
248
249     if actv in ('5740', '5745', '5741') then actv = '5750'; /*as per
Audrey Hu, 17Nov23*/
250     if actv <= '0900' and actv not in ('0350','0560') then actgrp=1;
251     else if '1020' <= actv <= '4910' or actv in ('0350','0560')
then actgrp=2;
252     else if '5300' <= actv <= '5750' then actgrp=3;
253     else if '5040' <= actv <= '6720' then actgrp=4;
SYMBOLGEN:  Macro variable FY resolves to 25
254     TITLE1 "FY 20&fy. Qtr A CARMM SYSTEM";
255 run;
```

```
NOTE: There were 102200 observations read from the data set WORK.B.
NOTE: The data set WORK.CARR has 102200 observations and 311 variables.
NOTE: DATA statement used (Total process time):
      real time          0.06 seconds
```

cpu time 0.04 seconds

```
256
257 proc sql;
258     create table smy as
259     select cag, fin, actv, bf, sum(dollar) as dollar
260     from carr
261     group by cag, fin, actv, bf;
NOTE: Table WORK.SMY created, with 411 rows and 5 columns.
```

```
262 quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.04 seconds
      cpu time           0.01 seconds
```

```
263
264 data smyx;
265     array bfx{4} bf1 - bf4;
266     do i=1 to 4;
267         set smy;
268         by cag fin actv bf;
269         if i=1 and bf='2' then i=i+1;
270         if i=1 and bf='3' then i=i+2;
271         if i=1 and bf='5' then i=i+3;
272         if i=2 and bf='3' then i=i+1;
273         if i=2 and bf='5' then i=i+2;
274         if i=3 and bf='5' then i=i+1;
275         bfx{i}=dollar;
276         if last.actv then i=4;
277     end;
278     if bf1=. then bf1=0;
279     if bf2=. then bf2=0;
280     if bf3=. then bf3=0;
281     if bf4=. then bf4=0;
282     total=sum(of bf1-bf4);
283     cagfin=cag !! fin;
284     if actv <= '0900' and actv not in ('0350','0560') then
actgrp=1;
285         else if '1020' <= actv <= '4910' or actv in ('0350','0560')
then actgrp=2;
286         else if '5300' <= actv <= '5750' then actgrp=3;
287         else if '5040' <= actv <= '6720' then actgrp=4;
288         output;
289 run;
```

```
NOTE: There were 411 observations read from the data set WORK.SMY.
NOTE: The data set WORK.SMYX has 400 observations and 13 variables.
NOTE: DATA statement used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds
```

```

290
291
292   proc sort data=smyx;
293       by cagfin actgrp actv;
294   run;

```

NOTE: There were 400 observations read from the data set WORK.SMYX.

NOTE: The data set WORK.SMYX has 400 observations and 13 variables.

NOTE: PROCEDURE SORT used (Total process time):

```

      real time          0.01 seconds
      cpu time           0.00 seconds

```

```

295   *-----;
296   *                                           ;
297   * Dollar tally summary by cag fin route actv bf. This      ;
298   * summary file is used in next step for distributing      ;
299   * mixed-mail.                                           ;
300   *                                           ;
301   *-----;
302

```

```

303   proc sql;
304       create table smy2 as
305       select cag, fin, route, actv, bf, sum(dollar) as dollar
306       from carr
307       group by cag, fin, route, actv, bf;

```

NOTE: Table WORK.SMY2 created, with 1575 rows and 6 columns.

```

308   quit;

```

NOTE: PROCEDURE SQL used (Total process time):

```

      real time          0.04 seconds
      cpu time           0.01 seconds

```

```

309
310
311   data smy2;
312       set smy2;
313
314   *Use macro to assign route group;
315       %assignRouteGroup;
MPRINT(ASSIGNROUTEGROUP):   *** The following statement assigns route
code groups *** so that
reports can be created separately for *** these groups.;
MPRINT(ASSIGNROUTEGROUP):   *set rgroup for special runs, for delivery
cost model (3 values);
MPRINT(ASSIGNROUTEGROUP):   *Put Letter Route types in RGroup 1, SPR in
RGroup 2, 99 in RGroup 3;
MPRINT(ASSIGNROUTEGROUP):   if '71' <= route <= '83' then RGroup=1;
MPRINT(ASSIGNROUTEGROUP):   else if '84' <= route <= '98' then RGroup=2;
MPRINT(ASSIGNROUTEGROUP):   else RGroup=3;
316       cagfin=cag !! fin;
317   run;

```

NOTE: There were 1575 observations read from the data set WORK.SMY2.  
NOTE: The data set WORK.SMY2 has 1575 observations and 8 variables.  
NOTE: DATA statement used (Total process time):  
    real time                0.01 seconds  
    cpu time                 0.00 seconds

```
318
319   data out;
320       array bff{4} dollar1-dollar4;
321       do i=1 to 4;
322           set smy2;
323           by cag fin route actv bf;
324               if i=1 and bf='2' then i=i+1;
325               if i=1 and bf='3' then i=i+2;
326               if i=1 and bf='5' then i=i+3;
327               if i=2 and bf='3' then i=i+1;
328               if i=2 and bf='5' then i=i+2;
329               if i=3 and bf='5' then i=i+1;
330               bff{i}=dollar;
331               if last.actv then i=4;
332       end;
333       if dollar1=. then dollar1=0;
334       if dollar2=. then dollar2=0;
335       if dollar3=. then dollar3=0;
336       if dollar4=. then dollar4=0;
337       drop i cag fin _type_ _freq_ dollar bf;
338   output;
339   run;
```

WARNING: The variable \_type\_ in the DROP, KEEP, or RENAME list has never been referenced.

WARNING: The variable \_freq\_ in the DROP, KEEP, or RENAME list has never been referenced.

NOTE: There were 1575 observations read from the data set WORK.SMY2.  
NOTE: The data set WORK.OUT has 1573 observations and 8 variables.  
NOTE: DATA statement used (Total process time):  
    real time                0.01 seconds  
    cpu time                 0.00 seconds

```
340
341
342   ****Note: this 2nd part of the program is analogous to the **;
343   ****      old program ALA860C0, which performed distribution **;
344   ****      and produced various summary reports.          **;
345   ****
346   ****Note: this 2nd part of the program is analogous to the **;
347   ****      old program ALA860C0, which performed distribution **;
348   ****      and produced various summary reports.          **;
349   ****
350   data out;
351       set out;
352       if '5300' <= actv <= '5750' then mixmail=1;
353       else if '1000' <= actv <= '5000' or actv in ('0350','0560')
354       then mixmail=0;
```



```
352      dollart=sum(of dollar1 - dollar4);
353  run;
```

NOTE: There were 1573 observations read from the data set WORK.OUT.

NOTE: The data set WORK.OUT has 1573 observations and 10 variables.

NOTE: DATA statement used (Total process time):

```
real time      0.02 seconds
cpu time       0.00 seconds
```

```
354
355      *-----;
356      * The following proc SUMMARY gets the preliminary results for ;
357      * Report 7 This corresponds to the old LIOCATT report ALA860P7;
358      *-----;
359  proc sort data=out out=out7;
360      by rgroup route actv;
361      where mixmail ne 1;
362  run;
```

NOTE: There were 1366 observations read from the data set WORK.OUT.

WHERE mixmail not = 1;

NOTE: The data set WORK.OUT7 has 1366 observations and 10 variables.

NOTE: PROCEDURE SORT used (Total process time):

```
real time      0.00 seconds
cpu time       0.00 seconds
```

```
363
364  proc  summary data=out7 noprint;
365      var dollar1 dollar2 dollar3 dollar4;
366      output out=rpt7 sum=;
367      by rgroup route actv;
368      /****uncomment print if report is needed***
369  proc print data=rpt7;
370      id rgroup route ;
371      * sum dollar1 dollar2 dollar3 dollar4;
372      by rgroup route;
373      title2 'Report 7';
374      *****/
375  run;
```

NOTE: There were 1366 observations read from the data set WORK.OUT7.

NOTE: The data set WORK.RPT7 has 303 observations and 9 variables.

NOTE: PROCEDURE SUMMARY used (Total process time):

```
real time      0.01 seconds
cpu time       0.00 seconds
```

```
376
377      *****;
378      * Distribute mixed mail costs to direct mail activity codes. ;
379      *****;
380
```

```

381      *-----;
382      * Read the mixed-mail to direct mail code conversion file.      ;
383      * This file tells which direct mail activity codes to associate;
384      * with each mixed-mail activity code.                          ;
385      *-----;
386      ;
387      Data mxmail;
388      set MxMailMap;
389      run;

```

NOTE: There were 293 observations read from the data set WORK.MXMAILMAP.

NOTE: The data set WORK.MXMAIL has 293 observations and 2 variables.

NOTE: DATA statement used (Total process time):

```

      real time          0.01 seconds
      cpu time           0.00 seconds

```

```

390
391      proc sort data=mxmail; by dircode;
392      run;

```

NOTE: There were 293 observations read from the data set WORK.MXMAIL.

NOTE: The data set WORK.MXMAIL has 293 observations and 2 variables.

NOTE: PROCEDURE SORT used (Total process time):

```

      real time          0.00 seconds
      cpu time           0.00 seconds

```

```

393
394      data dirmail (drop= i mixcode);
395          array dir(8) $mix1 -mix8;
396          do i=1 to 8;
397              set mxmail;
398              by dircode;
399              actv=dircode;
400              dir(i)=mixcode;
401              if last.dircode then i=8;
402          end;
403          output;
404      run;

```

NOTE: There were 293 observations read from the data set WORK.MXMAIL.

NOTE: The data set WORK.DIRMAIL has 101 observations and 10 variables.

NOTE: DATA statement used (Total process time):

```

      real time          0.01 seconds
      cpu time           0.00 seconds

```

```

405      *-----;
406      * Direct mail cost summary by cagfin, route and activity      ;
407      *-----;
408      proc sort data=out out=dist2;
409          by cagfin route actv;
410          where mixmail=0; * Contains only direct mail;

```

```
411 run;
```

NOTE: There were 946 observations read from the data set WORK.OUT.

WHERE mixmail=0;

NOTE: The data set WORK.DIST2 has 946 observations and 10 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time 0.01 seconds

cpu time 0.00 seconds

```
412
```

```
413 proc summary data=dist2 noprint;
```

```
414     var dollar1 dollar2 dollar3 dollar4 dollart;
```

```
415     output out=dirdol
```

```
416         sum=dollar1d dollar2d dollar3d dollar4d dollartd;
```

```
417     by cagfin route actv;
```

```
418     title2 'dirdol';
```

```
419     * proc print data=dirdol;
```

```
420 run;
```

NOTE: There were 946 observations read from the data set WORK.DIST2.

NOTE: The data set WORK.DIRDOL has 946 observations and 10 variables.

NOTE: PROCEDURE SUMMARY used (Total process time):

real time 0.01 seconds

cpu time 0.00 seconds

```
421
```

```
422     *-----;
```

```
423     * Merge direct mail cost with the conversion table to associate;
```

```
424     * each direct mail activity code with mixed mail code that will;
```

```
425     * get distribued cost.                                     ;
```

```
426     *-----;
```

```
427 proc sort data=dirdol;
```

```
428     by actv;
```

```
429 run;
```

NOTE: There were 946 observations read from the data set WORK.DIRDOL.

NOTE: The data set WORK.DIRDOL has 946 observations and 10 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time 0.00 seconds

cpu time 0.00 seconds

```
430 proc sort data=dirmail;
```

```
431     by actv;
```

```
432 run;
```

NOTE: There were 101 observations read from the data set WORK.DIRMAIL.

NOTE: The data set WORK.DIRMAIL has 101 observations and 10 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time 0.01 seconds

cpu time 0.00 seconds

```

433
434 data dircodel (drop=mix1-mix8 i);
435     merge dirdol (in=a)
436           dirmail (in=b keep=actv mix1-mix8);
437     by actv;
438     source=a+2*b;
439     if source=2 then delete; * delete activity codes that
contain
440                                     no cost;
441     array dir(8) mix1-mix8;
442     do i=1 to 8;
443         mixcode=dir(i);
444         if mixcode ne ' ' then output;
445         if mixcode= ' ' then i=8;
446     end;
447     *proc print n;
448 run;

```

NOTE: There were 946 observations read from the data set WORK.DIRDOL.  
NOTE: There were 101 observations read from the data set WORK.DIRMAIL.  
NOTE: The data set WORK.DIRCODE1 has 3033 observations and 12 variables.  
NOTE: DATA statement used (Total process time):  
real time 0.00 seconds  
cpu time 0.00 seconds

```

449
450 *-----;
451 * Mixed mail cost summary by cagfin, route and actv ;
452 *-----;
453
454 proc sort data=out out=dist1;
455     by cagfin route actv;
456     where mixmail=1; * Contains only mixed mails;
457 run;

```

NOTE: There were 207 observations read from the data set WORK.OUT.  
WHERE mixmail=1;  
NOTE: The data set WORK.DIST1 has 207 observations and 10 variables.  
NOTE: PROCEDURE SORT used (Total process time):  
real time 0.01 seconds  
cpu time 0.00 seconds

```

458
459 proc summary data=dist1 noprint;
460     var dollar1 dollar2 dollar3 dollar4 dollart;
461     output out=mxdol
462           sum=dollar1i dollar2i dollar3i dollar4i dollarti;
463     by cagfin route actv;
464     title2 'mxdol';
465 run;

```

NOTE: There were 207 observations read from the data set WORK.DIST1.  
NOTE: The data set WORK.MXDOL has 207 observations and 10 variables.  
NOTE: PROCEDURE SUMMARY used (Total process time):  
    real time            0.01 seconds  
    cpu time             0.00 seconds

```
466
467      *-----;
468      * Attach summary mixed mail cost to the direct mail activity ;
469      * code. ;
470      *-----;
471
472      data mxdol;
473          set mxdol;
474          rename actv=mixcode;
475          *proc print data=mxdol;
476          run;
```

NOTE: There were 207 observations read from the data set WORK.MXDOL.  
NOTE: The data set WORK.MXDOL has 207 observations and 10 variables.  
NOTE: DATA statement used (Total process time):  
    real time            0.01 seconds  
    cpu time             0.00 seconds

```
477
478      proc sort data=dircode1;
479          by cagfin route mixcode;
480          run;
```

NOTE: There were 3033 observations read from the data set WORK.DIRCODE1.  
NOTE: The data set WORK.DIRCODE1 has 3033 observations and 12 variables.  
NOTE: PROCEDURE SORT used (Total process time):  
    real time            0.01 seconds  
    cpu time             0.00 seconds

```
481      proc sort data=mxdol;
482          by cagfin route mixcode;
483          run;
```

NOTE: There were 207 observations read from the data set WORK.MXDOL.  
NOTE: The data set WORK.MXDOL has 207 observations and 10 variables.  
NOTE: PROCEDURE SORT used (Total process time):  
    real time            0.00 seconds  
    cpu time             0.00 seconds

```
484
485      data final ;
486          merge dircode1 (in=a) mxdol (in=b);
487          by cagfin route mixcode;
488          source=a+2*b;
```

489  
490  
491  
492

```
* proc print;  
* title2 'final';  
run;
```

NOTE: There were 3033 observations read from the data set WORK.DIRCODE1.  
NOTE: There were 207 observations read from the data set WORK.MXDOL.  
NOTE: The data set WORK.FINAL has 3033 observations and 17 variables.  
NOTE: DATA statement used (Total process time):  
real time 0.01 seconds  
cpu time 0.00 seconds

493  
494  
495  
496  
497  
498

```
*-----;  
* Calculate the total direct mail cost associated with ;  
* each mixed mail code. ;  
*-----;
```

```
499 proc sort data=final;  
500     by cagfin route mixcode;  
501     run;
```

NOTE: There were 3033 observations read from the data set WORK.FINAL.  
NOTE: The data set WORK.FINAL has 3033 observations and 17 variables.  
NOTE: PROCEDURE SORT used (Total process time):  
real time 0.01 seconds  
cpu time 0.00 seconds

502  
503  
504  
505  
506  
507  
508

```
proc summary data=final noprint;  
    var dollar1d dollar2d dollar3d dollar4d dollartd;  
    output out=total1  
           sum=total1d total2d total3d total4d totaltd;  
    by cagfin route mixcode;  
run;
```

NOTE: There were 3033 observations read from the data set WORK.FINAL.  
NOTE: The data set WORK.TOTAL1 has 413 observations and 10 variables.  
NOTE: PROCEDURE SUMMARY used (Total process time):  
real time 0.01 seconds  
cpu time 0.01 seconds

509  
510  
511  
512  
513  
514  
515  
516  
517

```
*-----;  
* Distribute the mixed mail cost to direct mail activity codes ;  
*-----;  
data distfile (drop=_type_ _freq_ source);  
    merge final (in=a) total1 (in=b);  
    by cagfin route mixcode;  
    if total1d=0 then ratio1=.;
```

```

518         else ratio1= dollar1d/total1d;
519         dist_1=dollar1i *ratio1;
520
521         if total2d=0 then ratio2=.;
522         else ratio2= dollar2d/total2d;
523         dist_2=dollar2i *ratio2;
524
525         if total3d=0 then ratio3=.;
526         else ratio3= dollar3d/total3d;
527         dist_3=dollar3i *ratio3;
528
529     *** The following statements add the undistributed dollar
530     *** for basic functions 1 - 3 to last basic function;
531
532         if (total1d =0 and dollar1i > 0)
533         then dollar4i=dollar4i +dollar1i;
534         if (total2d =0 and dollar2i > 0)
535         then dollar4i=dollar4i +dollar2i;
536         if (total3d =0 and dollar3i > 0)
537         then dollar4i=dollar4i +dollar3i;
538
539     *** Note that the last basic function distribution is based on
540     *** the total cost (sum of four basic functions cost);
541
542         if totaltd=0 then ratiot=.;
543         else ratiot= dollartd/totaltd;
544         dist_4=dollar4i *ratiot;
545
546     *** The following code assigns the undistributed mixed
547     *** dollars to the dist_0 cagegory;
548
549         if actv=' ' then
550             do;
551                 actv=mixcode;
552                 dist_0=dollarti;
553             end;
554
555     *Use macro to assign route group;
556     %assignRouteGroup;
557 MPRINT(ASSIGNROUTEGROUP):  *** The following statement assigns route
558 code groups *** so that
559 reports can be created separately for *** these groups.;
560 MPRINT(ASSIGNROUTEGROUP):  *set rgroup for special runs, for delivery
cost model (3 values);
561 MPRINT(ASSIGNROUTEGROUP):  *Put Letter Route types in RGroup 1, SPR in
RGroup 2, 99 in RGroup 3;
562 MPRINT(ASSIGNROUTEGROUP):  if '71' <= route <= '83' then RGroup=1;
563 MPRINT(ASSIGNROUTEGROUP):  else if '84' <= route <= '98' then RGroup=2;
564 MPRINT(ASSIGNROUTEGROUP):  else RGroup=3;
565
566     * proc print data=distfile n;
567 run;

```

NOTE: Missing values were generated as a result of performing an operation on missing values.

Each place is given by: (Number of times) at (Line):(Column).

2940 at 519:27    687 at 523:27    3033 at 527:27    687 at 544:27

NOTE: There were 3033 observations read from the data set WORK.FINAL.

NOTE: There were 413 observations read from the data set WORK.TOTAL1.

NOTE: The data set WORK.DISTFILE has 3033 observations and 29 variables.

NOTE: DATA statement used (Total process time):

real time            0.04 seconds

cpu time            0.00 seconds

```
561
562      *-----;
563      * Combine schedule K and L to produce K and L report      ;
564      * (report 13). Schedule K is report 7 and schedule L is    ;
565      * summary of the mixed mail distribution result (finout).  ;
566      *-----;
567
568      proc sort data=distfile;
569          by   rgroup route actv;
570          run;
```

NOTE: There were 3033 observations read from the data set WORK.DISTFILE.

NOTE: The data set WORK.DISTFILE has 3033 observations and 29 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time            0.00 seconds

cpu time            0.00 seconds

```
571
572      proc summary data=distfile noprint;
573          var   dist_0 dist_1 dist_2 dist_3 dist_4;
574          output out=finout1 sum=dollar0 dollar1 dollar2 dollar3
dollar4;
575          by   rgroup route actv;
576          run;
```

NOTE: There were 3033 observations read from the data set WORK.DISTFILE.

NOTE: The data set WORK.FINOUTL has 213 observations and 10 variables.

NOTE: PROCEDURE SUMMARY used (Total process time):

real time            0.01 seconds

cpu time            0.00 seconds

```
577
578      *-----;
579      * The following proc format classifies activity codes into ;
580      * categories for subsequent reports.                        ;
581      *-----;
582      proc format ;
583          value $actv
584          /*SPECIAL SERVICES */
585              '0060'          ='055 Registered  '
```



```

586          '0050'          ='051 Certified    '
587          '0080'          ='054 Insured      '
588          '0030'          ='052 COD          '
589
590      /* OTHSERV INCL. BUSRPLY, RETRCPT, ADDRRCORR, FORMS3547&3579 */
591          '6020','6030'='074 PO Box      '
592          '0090','0190',
593          '0300','0100',
594          '0110','0140','0200'          ='058 Other Services' /*17Nov23
add 0200 Label
594! Delivery*/
595
596      /* FIRST CLASS      */
597          '1060'          ='003 FC SP Letters  '
598          '1020'          ='004 FC SP Cards    '
599          '1080','1081','1085','1086'='008 FC Prst Letters'
600          '1022','1035','1040','1045'='009 FC Prst Cards  '
601          '2060'          ='014 FC SP Flats    '
602          '2080','2081','2086'          ='017 FC Prst Flats  '
603          '3060','4060'          ='143 FC Package Svc  '
604          /*'3080','4080'          ='020 FC Prst Parcels'*/
605          '3080','4080'          ='143 FC Package Svc  '
606
607
608      /*PRIORITY MAIL      */
609          '1160','2160','3160','4160'='148 Priority      '
610
611      /* EXPRESS MAIL      */
612          '1110','2110','3110','4110'='140 Express      '
613
614      /* PERIODICALS      */
615          '1211','2211','3211','4211'='031 PER In County  '
616          '1212','2212','3212','4212'='032 PER Outside County'
617
618      /* STANDARD      */
619          '1317'          ='021 MM ECR HD Letters
620          '2317','3317','4317'          ='022 MM ECR HD Flats/Parcels
621          '1318'          ='021 MM ECR SAT Letters
622          '2318','3318','4318'          ='022 MM ECR SAT
Flats/Parcels  '
623          '1312','2312','3312','4312'='023 MM ECR Carrier Route
624          '1320','2320','3320','4320'='024 MM EDDM - Retail
625          '1340','1341','1345'          ='025 MM REG Letters
626          '2340','2341','2345'          ='026 MM REG Flats
627          '3340','4340'          ='027 MM REG Parcel

```

```

628          '3341','4341'                ='027 MM REG Parcel
629          '1315'                        ='021 DAL Ltr WSH
630          '2315','3315','4315'         ='021 DAL Flt/IPP/PAR WSH
631          '1316'                        ='022 DAL Ltr WSS
632          '2316','3316','4316'         ='022 DAL Flt/IPP/PAR WSS
633          '1319','2319','3319','4319'  ='021 Parent No Dal HD/SAT
634
635  /* PACKAGE                          */
636          '2402','3402','4402'          ='157 Ground Advantage HW 1-
31b '
637          '2403','3403','4403'          ='158 Ground Advantage HW
over31b '
638          '2405','3405','4405'          ='156 Ground Advantage LW
639          '2410','3410','4410'          ='145 PKG Retail Ground
640          '2460','2465','2480','2495'  ='042 PKG BPM Flats
641          '3460','3465','3480','3495',
642          '4460','4465','4480','4495'  ='043 PKG BPM Parcels
643          '1420','1425','1430',
644          '2420','2425','2430',
645          '3420','3425','3430',
646          '4420','4425','4430'          ='044 PKG Media
647
648  /* PARCEL SELECT and PS Lightweight    */
649          '2360','3360','4360',
650          '2493','3493','4493'          ='151 PKG Parcel Select
651
652  /* PARCEL RETURN SERVICE (PRS)          */
653          '2475','3475','4475'          ='154 PKG Parcel Return (PRS)
654
655  /* PREMIUM FORWARDING SERVICE (PFS)    */
656          '0350','1170','2170','3170','4170'='170 Prem. Forw. Svc'
657
658
659  /* GOVERNMENT MAIL                      */
660          '0560',
661          '1510','2510','3510','4510'  ='085 USPS
662
663  /* FREE                                  */
664          '1910','2910','3910','4910'  ='086 Free
665
666  /* INTERNATIONAL OUTBOUND AND INBOUND  */

```

```

667          '1780','2780','3780','4780',
668          '5081','6081','5082','6082',
669          '0752','0757','0762','0767',
670          '0770','0860','0862','0867'           ='185

```

International'

```

671
672  /* OTHER CODES */
673      OTHER = '999 OTHER';

```

NOTE: Format \$ACTV has been output.

```

674  run;

```

NOTE: PROCEDURE FORMAT used (Total process time):

real time                0.06 seconds

cpu time                 0.00 seconds

```

675
676
677  data KL (drop=_type_ _freq_);
678      set rpt7 (in=a) finout1 (in=b);
679      if a then source='K';
680      if b then source='L';
681      if dollar0 > 0 then
682          do;   source='U';
683              dollar4=dollar0; *move the undistributed mixed
mail
684                          cost to basic function OTHER;
685          end;
686
687      if dollar0=. and dollar1=. and dollar2=. and dollar3=. and
688      dollar4=. then delete;
689
690      total= sum (of dollar1 - dollar4);
691      if actv <= '0900' and actv not in ('0350','0560')
then actgrp=1;
692      else if '1020' <= actv <= '4910' or actv in
('0350','0560') then actgrp=2;
693      else if '5300' <= actv <= '5750' then actgrp=3;
694      else if '5020' <= actv <= '6720' then actgrp=4; **bls;
695      **use format to classify activity to classes**;
696      class=put(actv,$actv.);
697
698      **assign labels for use in reports**;
699      label
700          dollar1 = 'OUTGOING'
701          dollar2 = 'INCOMING'
702          dollar3 = 'TRANSIT '
703          dollar4 = 'OTHER   ';
704
705                      **restrict to letter routes only**;
706                      *if rgroup=1;
707  run;

```

NOTE: There were 303 observations read from the data set WORK.RPT7.

NOTE: There were 213 observations read from the data set WORK.FINOUTL.  
NOTE: The data set WORK.KL has 515 observations and 12 variables.  
NOTE: DATA statement used (Total process time):  
    real time                0.02 seconds  
    cpu time                  0.00 seconds

```
708
709
710 *Generate class variable from lookup table;
711 *import craft,activity code combinations used in delivery cost
model;
712 PROC IMPORT OUT= WORK.ActivityCodesTable
713             DATAFILE=
"&pathprog\ActivityCodesMaster_FilingFY25.xlsx"
SYMBOLGEN:  Macro variable PATHPROG resolves to D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles
714             DBMS=XLSX REPLACE;
715 RUN;
```

NOTE: One or more variables were converted because the data type is not supported by the V9 engine. For more details, run with options MSGLEVEL=I.  
NOTE: The import data set has 163 observations and 6 variables.  
NOTE: WORK.ACTIVITYCODESTABLE data set was successfully created.  
NOTE: PROCEDURE IMPORT used (Total process time):  
    real time                0.05 seconds  
    cpu time                  0.00 seconds

```
716 proc sql;
717 create table KL2 as
718     select KL.*, ac.NewCRAProd as class2
719     from KL left join ActivityCodesTable as ac
720     on KL.actv = ac.ActivityCode;
NOTE: Table WORK.KL2 created, with 515 rows and 13 columns.
```

```
721 quit;
NOTE: PROCEDURE SQL used (Total process time):
    real time                0.01 seconds
    cpu time                  0.00 seconds
```

```
722
723 data checkClass; set KL2;
724 where class ne class2;
725 run;
```

NOTE: There were 192 observations read from the data set WORK.KL2.  
    WHERE class not = class2;  
NOTE: The data set WORK.CHECKCLASS has 192 observations and 13 variables.  
NOTE: DATA statement used (Total process time):  
    real time                0.00 seconds  
    cpu time                  0.00 seconds

```

726
727 ***zzz;
728
729 data KL; set KL2;
730 if class2 = "" then class2 = "999 OTHER";
731 if substr(class2,1,3) = "185" then class = "185 International";
732     else class = class2;
733 drop class2;
734 run;

```

NOTE: There were 515 observations read from the data set WORK.KL2.

NOTE: The data set WORK.KL has 515 observations and 12 variables.

NOTE: DATA statement used (Total process time):

|           |              |
|-----------|--------------|
| real time | 0.00 seconds |
| cpu time  | 0.00 seconds |

```

735
736 proc sort data=kl;
737 by rgroup route actv;
738 run;

```

NOTE: There were 515 observations read from the data set WORK.KL.

NOTE: The data set WORK.KL has 515 observations and 12 variables.

NOTE: PROCEDURE SORT used (Total process time):

|           |              |
|-----------|--------------|
| real time | 0.01 seconds |
| cpu time  | 0.00 seconds |

```

739
740 *Export KL table by route, for delivery cost model;
741     proc export data=kl
742         outfile = "&pathOut\KLTable&FYData..xlsx"
SYMBOLGEN:  Macro variable PATHOUT resolves to D:\ExternalSSD\Folder
19\FY25\SAS\KLTables
SYMBOLGEN:  Macro variable FYDATA resolves to 25
743         dbms = xlsx replace;
744 run;

```

NOTE: The export data set has 515 observations and 12 variables.

NOTE: "D:\ExternalSSD\Folder 19\FY25\SAS\KLTables\KLTable25.xlsx" file was successfully created.

NOTE: PROCEDURE EXPORT used (Total process time):

|           |              |
|-----------|--------------|
| real time | 0.04 seconds |
| cpu time  | 0.00 seconds |